SMART INDIA HACKATHON 2026

TITLE PAGE

Problem Statement ID – SIH26056

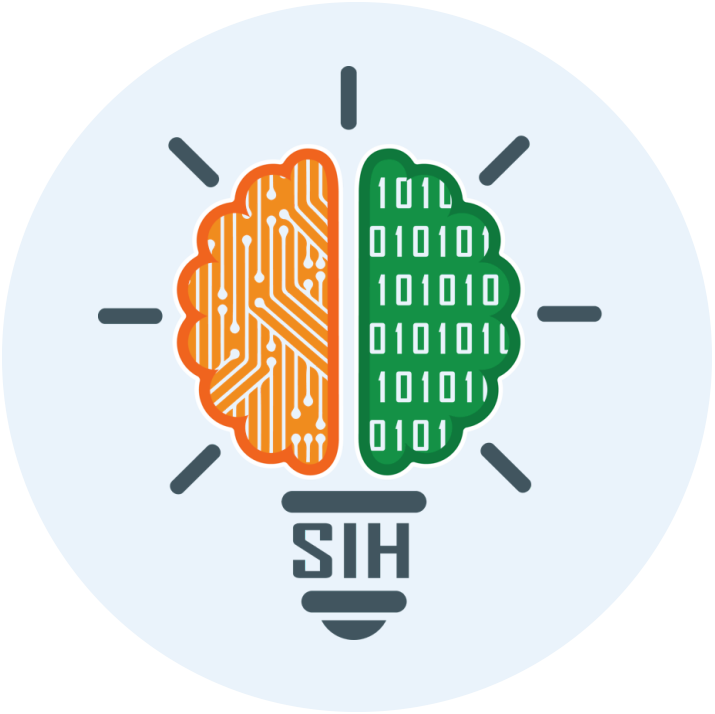
Problem Statement Title – Development of a Real-time Airfare Price Index for India through Automated Web Scraping of Airline and Online Travel Aggregator Portals for Augmentation of the Consumer Price Index (CPI)

Theme – Travel & Tourism

PS Category – Software

Team ID – _____

Team Name (Registered on portal) – AeroDex



Ministry of Statistics and Programme Implementation (MoSPI) · Data Informatics & Innovation Division

AeroDex

A reproducible, daily Domestic Airfare Price Index (DAPI) for India, built to national-statistics standards.

- 60 routes · 420 strata
- 183 automated tests passing
- Bit-identical re-runs (SHA-256)
- Refuses to publish synthetic data

AeroDex

AeroDex — A Daily Airfare Price Index for India

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Proposed Solution

AeroDex prices a fixed panel of 60 domestic routes at 7 booking horizons, three times a day, and publishes a daily, DGCA-weighted airfare index that anyone can recompute from its published inputs — giving the CPI a sub-monthly, route-level signal it does not have today.

THE MEASUREMENT GAP

	CPI today	AeroDex
Cadence	Monthly	**Daily · 3 IST slots**
Look-ahead	One spec	**7 horizons, 1–60 days**
Coverage	Sampled	**60 routes · 23 airports**
Disclosure	Index level	**+ coverage, imputation, hash**

HOW IT WORKS

1 Observe
- 60 routes × 7 horizons
- 3 fixed IST slots a day
- All-in fare, as a traveller sees it

2 Clean
- Normalise, dedup, tag attributes
- Index the all-inclusive fare, as paid
- Bounds checks, quarantine

3 Index
- Matched-model Jevons per stratum
- Lowe aggregation, DGCA weights
- Missing data imputed, capped at 5%

4 Publish
- Dashboard and open API
- Coverage, imputation, hashes on every point
- aerodex verify re-derives it

INNOVATION AND UNIQUENESS

Reproducible by construction

A pure function: the same archived panel and config always give the same SHA-256. CI re-checks it every night.

Proof · aerodex verify → REPRODUCIBLE

Refuses to overstate

Blocks synthetic-only runs and any run over 5% imputed weight. Coverage failures are published, never hidden.

Proof · PUBLICATION REFUSED, exit code 3

Horizon-controlled

Fixed advance-purchase windows separate a real fare rise from late booking, as the IMF TA to MoSPI advised.

Proof · 7 horizons in config/panel.yaml

Statistical-office method

Jevons + Lowe, the family MoSPI's CPI 2024 uses, on DGCA 2025 traffic weights for all 60 routes.

Proof · 60 of 60 DGCA route weights

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TECHNICAL APPROACH

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End-to-end pipeline · status as of 24 Sep 2026

DATA SOURCES · 3-TIER LADDER

T1	**Institutional feed** MoSPI / DGCA route	PLANNED
T2	**Licensed fare API** HasData / SerpApi · Duffel	NEXT
T3	**Fixture, demo only** Never publishable	BUILT

1	2	3	4	5	6	7
Schedule	**Acquire**	**Normalise**	**Validate**	**Store**	**Index**	**Publish**
systemd timers, 3 IST slots ±15 min	ladder + robots.txt and rate-limit guard	dedup, attribute tags, all-in fare	plausibility bounds, quarantine	append-only quotes, DB triggers	pure function, Jevons then Lowe	refusal gate, JSON artifacts
BUILT	FIXTURE	BUILT	BUILT	BUILT	BUILT	BUILT

PostgreSQL + TimescaleDB ▸ static JSON artifacts ▸ FastAPI (11 endpoints) ▸ Next.js dashboard (9 screens)

TECHNOLOGIES TO BE USED

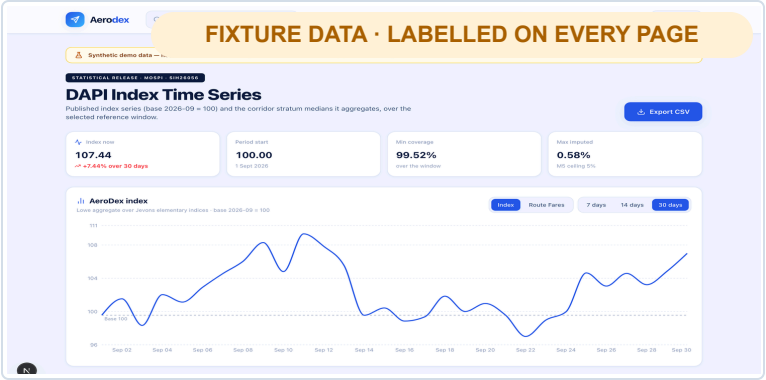
Language, data	Python 3.12 · pandas · NumPy · Pydantic
Storage	PostgreSQL 16 · TimescaleDB · append-only triggers
Collection	httpx · own RFC 9309 robots parser · systemd timers
Serving	FastAPI · Next.js 16 · React 19 · TypeScript · Recharts
Assurance	pytest (183) · ruff · mypy · GitHub Actions, nightly hash check
Hosting	Vercel (API live) · Cloudflare R2 + Oracle A1 as the ₹0 target

INDEX METHODOLOGY

1	Stratify route × horizon → 420 strata	BUILT
2	Matched-model Jevons per stratum	BUILT
3	Lowe aggregation, DGCA 2025 weights	BUILT
4	Group-mean imputation, capped at 5%	BUILT
5	Chained daily series, 2026-09 = 100	BUILT
6	Hedonic quality adjustment (log-fare)	NEXT
7	Seasonal adjustment (STL / X-13)	PLANNED

Every point ships: value · coverage · imputed share · n quotes · config hash · panel hash

WORKING PROTOTYPE




```
$ aerodex verify
weighted   c3523b87…ba27   OK
REPRODUCIBLE
```

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FEASIBILITY AND VIABILITY

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PS SIH26056 deliverables — where we stand

BUILT — **Cleaned fare database with metadata**
Postgres / Timescale schema, append-only quote archive, normalise, validate, quarantine.

BUILT — **Index-construction module**
Matched-model Jevons to Lowe on DGCA 2025 weights; imputation gate; 11 golden tests.

BUILT — **Interactive daily-index dashboard**
9 screens over an 11-endpoint API; every page labels fixture data.

BUILT — **Documentation and testing**
183 tests passing; API, data-dictionary, operations and methodology docs.

PARTIAL — **Multi-source collection engine (Python)**
Adapter framework, robots and rate-limit guard, 3-tier ladder done; live adapter is next.

PARTIAL — **Fare split: base, taxes, fees**
Dashboard shows a labelled estimate from statutory rates; the index uses the all-inclusive fare.

PARTIAL — **30-day back-test against DGCA data**
Harness runs on the fixture panel; a valid back-test needs 30 days of real quotes.

Challenges and risks → how we overcome them

No lawful free fare feed
OTA terms bar automation; Amadeus self-service closed 17 Jul 2026. Licensed fare-search API for the MVP; MoSPI / DGCA route pursued in parallel.

A source blocks or changes
Drop it and publish the coverage ratio. Imputation is capped at 5% or the run is refused.

API quota and cost
1,260 stratum-slots a day (~1,155 after overlap suppression), sized to a ~$99/month plan.

Fare attributes missing
Hedonic set cut to the fields the source returns; matched-model comparison stays in force.

Fixture read as a measurement
data_source tag on every API response, banner on every page, refusal in the publisher.

PS lists 5 airlines and 6 OTAs. The fixture models all 5 airlines; OTA pages are not scraped, because their terms bar automated access.

VIABILITY

Cost — A 30-day demo run ≈ $25; the full 60-route panel ≈ $99/month (provider pages, 25 Aug 2026).

Handover — A ₹0-recurring configuration on free tiers is kept for a ministry handover.

Roadmap — Now: pipeline + dashboard. Next: licensed adapter, 30 real days. Later: hedonic, seasonal, institutional feed.

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IMPACT AND BENEFITS

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Potential impact on the target audience

MoSPI
A daily, route-level series beneath the monthly CPI airfare item, with coverage and imputation disclosed for every point.

DGCA
Tariff spot checks become a continuous, horizon-controlled series: which routes move, how early, and why.

Travellers and researchers
Open dashboard, API and method: what a festival window costs, how far ahead to book, LCC against full-service.

The wider statistical system
Append-only archive, hashed config, pure function and golden tests are reusable for any high-frequency price.

Benefits of the solution

Social
- Trust through a disclosed method and open code
- Better-informed travel decisions
- A price people feel, measured in public

Economic
- 30-day demo ≈ $25; full panel ≈ $99/month
- ₹0-recurring handover configuration
- Earlier read on transport prices for policy

Technological
- Open API and a reproducible pipeline pattern
- Pure-function index, golden tests, hash verify
- Same engine extends to other high-frequency prices

Environmental
- Digital collection replaces field visits for this item
- Sized for one small VM: 2 OCPU, 12 GB
- A modest benefit, stated plainly

INDEPENDENTLY CORROBORATED BY MoSPI's OWN RECORD

CPI 2024 FAQ, Q20–21 — Jevons for elementary indices, Young / modified Laspeyres above — the same family AeroDex uses.

IMF TA, Recommendation 11 — Wider advance-purchase timings (14 and 21 days) and the airline as a price-determining characteristic.

Expert Group, Rec. 11(a) — DGCA already holds min / max tariff data for all routes — a documented institutional route to data.

SDG 17.18 · timely, reliable data | **SDG 9 · innovation, infrastructure** | **Digital India** | **Atmanirbhar Bharat · open source**

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RESEARCH AND REFERENCES

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How others build an airfare index — and where AeroDex sits

Who	How fares are obtained	Cadence	Lesson for AeroDex
China CAAC · TravelSky	A state reservation system sees every domestic fare; the regulator sets fare bands.	Monthly ticket-price index	Owns the rails: India needs a data mandate.
United States BLS · DOT	A mandatory ticket sample (14 CFR 241) selects routes; BLS prices them from a fare database.	Monthly CPI	Sample routes from real tickets; fixed advance-purchase specs.
UK · Australia Netherlands	The statistics office runs its own web-scraping unit on a fixed route panel.	Monthly, priced N days ahead	Fixed panel and fixed windows: AeroDex's design.
India today	CPI 2024 collects airfare from online platforms; airlines refused DGCA's passenger-level data request (Mar 2026).	Monthly	The gap PS SIH26056 targets.
Typical scraper prototype	Direct OTA scraping against site terms; breaks when the page layout changes.	Ad hoc	Not reproducible or defensible as an official statistic.
AeroDex	Licensed fare API now, institutional feed as the target; fixed 60 × 7 panel.	Daily, 3 IST slots	**A reproducible bridge to a mandate-based feed.**

OUR RESEARCH RECORD (docs/research/ in the repository)

6	research passes	**40+**	sources reviewed	**20 × 29**	providers × axes	**3**	verdicts revised, all kept

References

[1] MoSPI, FAQs on CPI 2024 series (Q20–21, Q27) — mospi.gov.in

[2] MoSPI, Expert Group Report on CPI updation (IMF TA Rec. 11; Rec. 11(a)) — mospi.gov.in

[3] ILO / IMF / OECD / UNECE / World Bank, Consumer Price Index Manual: Concepts and Methods

[4] Eurostat, Practical guidelines on web scraping for the HICP (2020)

[5] US BLS, CPI airline fares — bls.gov; DOT O&D Survey, 14 CFR Part 241

[6] UK ONS and ABS, web scraping in consumer price statistics

[7] DGCA city-pair traffic — Vonter/india-aviation-traffic (ODbL)

[8] Collection of Statistics Act, 2008 — mospi.gov.in

[9] RFC 9309, Robots Exclusion Protocol — rfc-editor.org

[10] MoSPI, CPI: changes in the revised series (base 2012 = 100) — cpi.mospi.gov.in

[11] Code, tests and decision log — github.com/Naveen-Boddepalli/aerodex